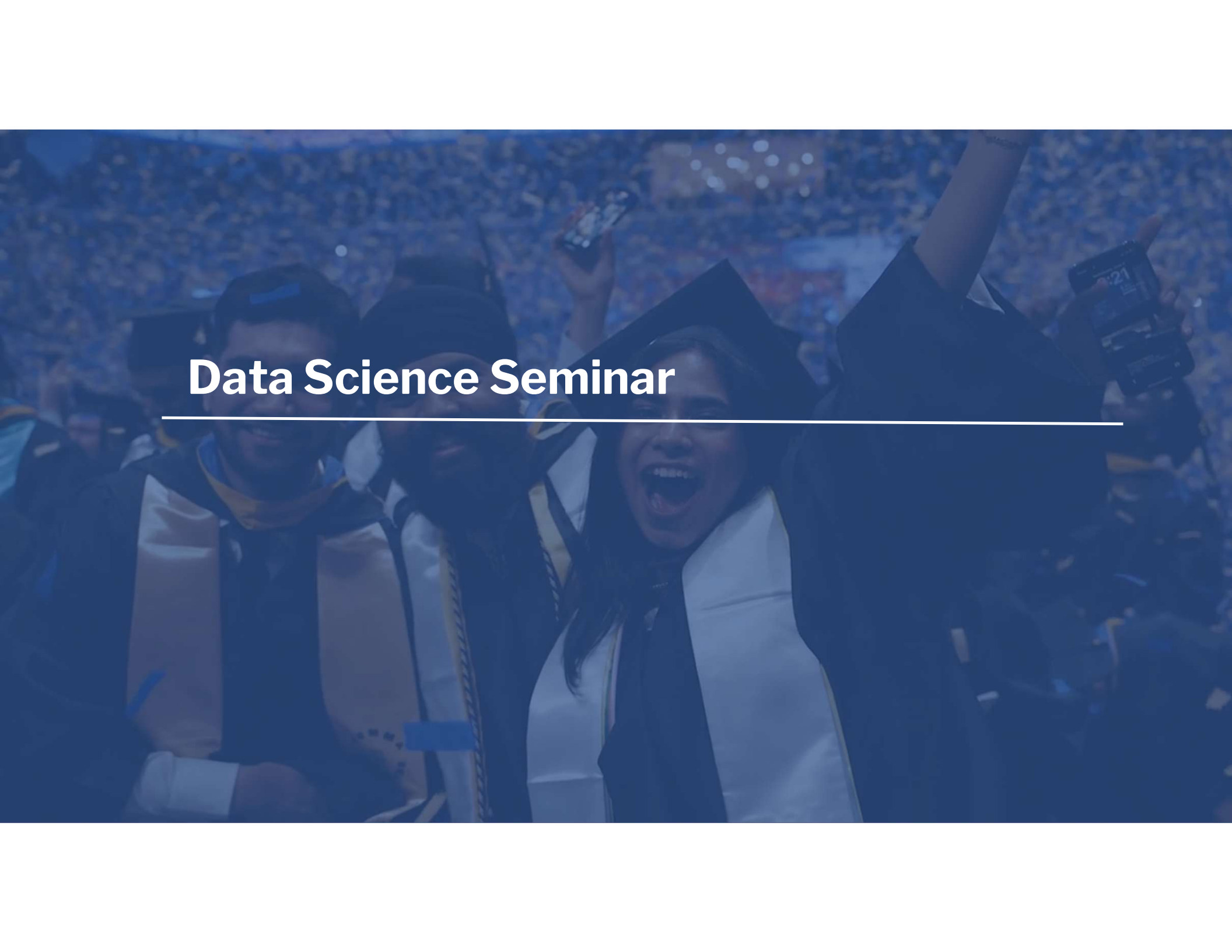


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large blue rectangular box is centered over the image, containing the text 'WEEK 3'.

Introduction to Generative AI (GAI 602)

WEEK 3

A photograph of graduates in black gowns and white stoles celebrating at night. They are holding up their phones to capture the moment, and confetti is visible in the air. The image has a blue color overlay.

Data Science Seminar

CUNY SPS Data Science Seminar

Dr. Adam Kelleher
Columbia University



Understanding and Using Causal Models for Forecasting

Causal Inference tells us what will happen if we make an intervention in a real world system. We will introduce the basic idea of Causality and discuss how to train Causal Models. Then we will discuss several real world case studies from a variety of different industries where using a Causal Model instead of a predictive one made a large difference to operations.

Thursday, October 1st
6:30PM
119th West 31st Street
Room 405

Sign up link (please sign up even if you are coming in person, you can put whatever you want for your emplid and the form will take it)
<https://us02web.zoom.us/j/83840080608?pwd=KwzeAlG66KJQ27QeAqBGkqkr5Xsxml.1&jst=1>



A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed upper section and a clock face is visible. To the right, a modern building with a glass facade is partially obscured by trees. In the foreground, a wide, paved walkway leads towards the buildings, with several students walking. On the left, a student is sitting on a wooden bench, looking at a phone. A black lamppost stands near the bench. The overall scene is a typical university setting with greenery and academic buildings.

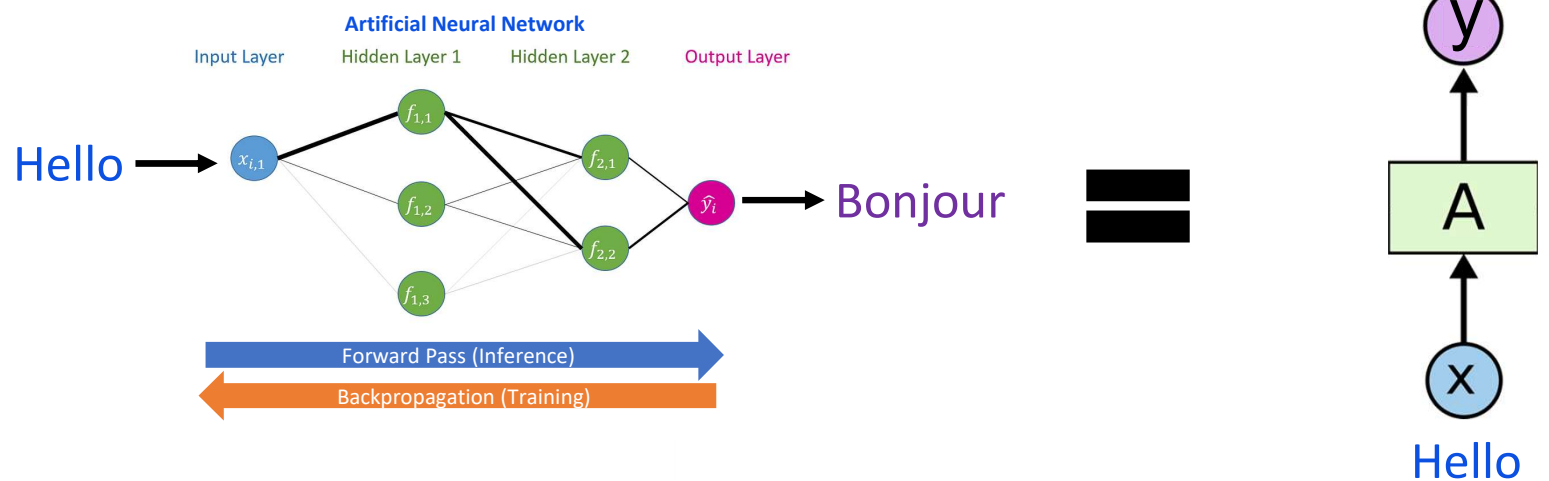
Sequence to Sequence Models

Use-cases for Seq-2-Seq models

- **Named entity recognition**
- **Natural language generation**
- **Music composition**
- **Handwriting recognition**
- **Language identification**
- **Paraphrase detection**
- **Speech recognition**

Sequence to Sequence Models

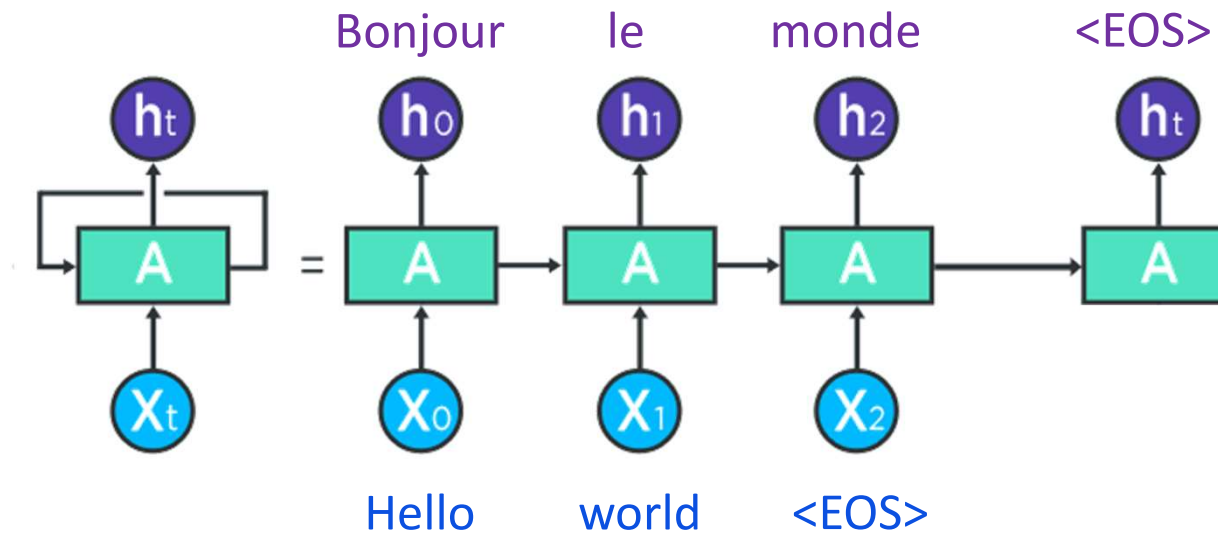
Let's say we want to translate “Hello world” into French.



Recurring Neural Network (RNN)

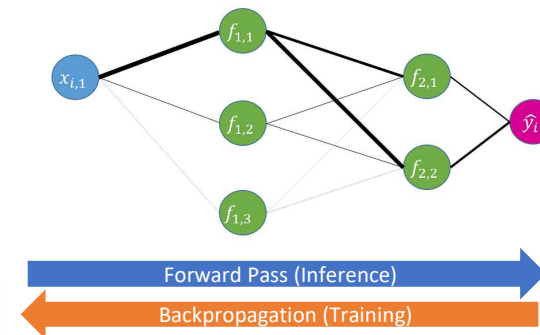
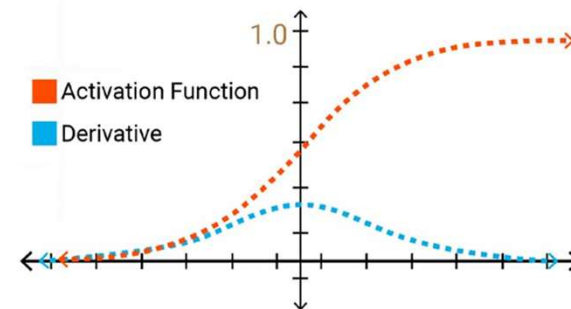
We need a way to remember the **context** to help generate the right word.

RNNs support the concept of **memory**, supporting data sequences



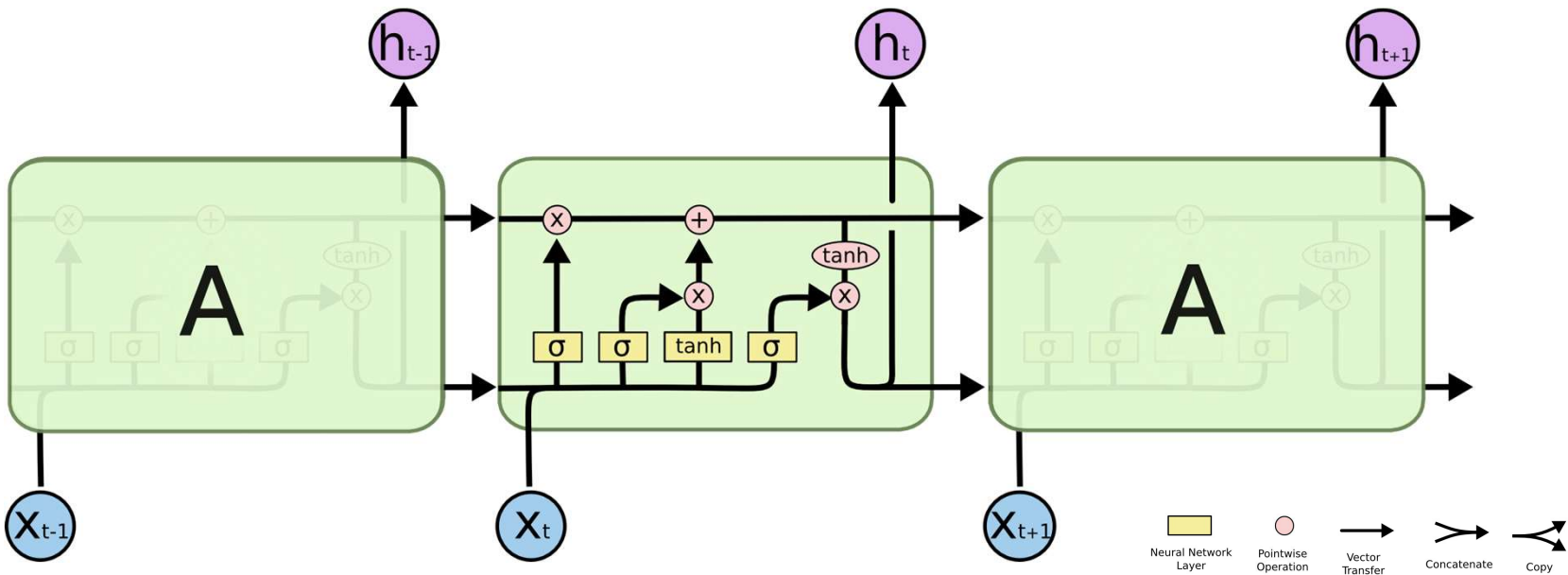
Vanishing Gradient Problem

- Deep neural networks use backpropagation.
 - Back propagation uses the chain rule.
 - The chain rule multiplies derivatives.
 - Often these derivatives between 0 and 1.
As the chain gets longer, products get smaller until they disappear
- or...
- As the chain gets longer, products get larger until they explode



Long short-term memory (LSTM) model

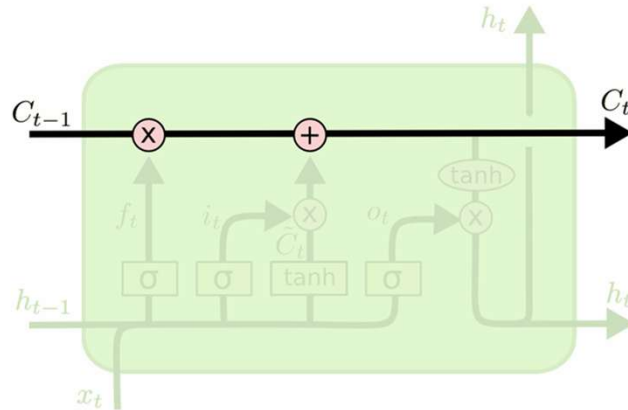
- LSTM uses **memory cells** to store information at each time step
- Uses **gates** to control the flow of information through the network
- How it works:
 - **Forget gate**: limit information passed from one cell to the next
 - **Input gate**: protect the current step from irrelevant inputs
 - **Output gate**: prevent the current step from passing irrelevant outputs to later steps



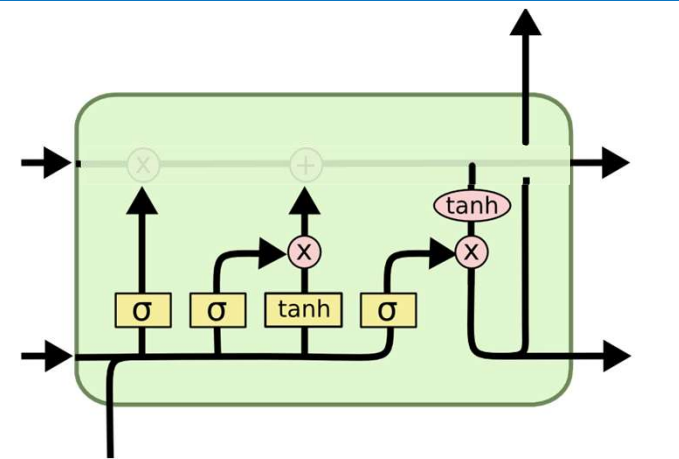
Source: <http://colah.github.io/posts/2015-08-Understanding-LSTMs>

LSTM: Key Concepts

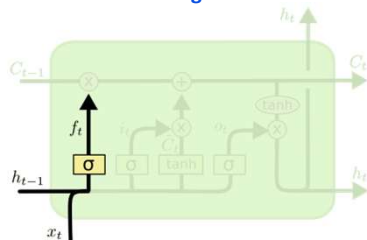
A **Backbone** to carry state forward and gradients backward



Gating to modulate information flow

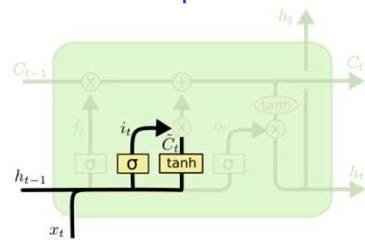


Forget



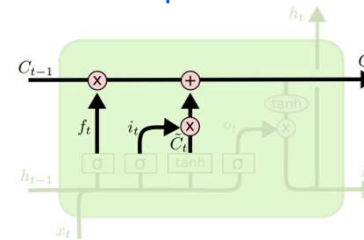
Use previous state, C , previous output, h , and current input, x , to determine how much to suppress previous state

Input



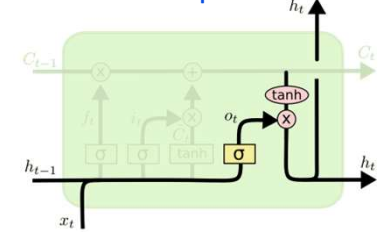
Input gate i determines which values of C to update. Separate tanh layer produces new state to add to C

Update C



Forget gate does attenuates C , and then adds output of previous input step.

Output

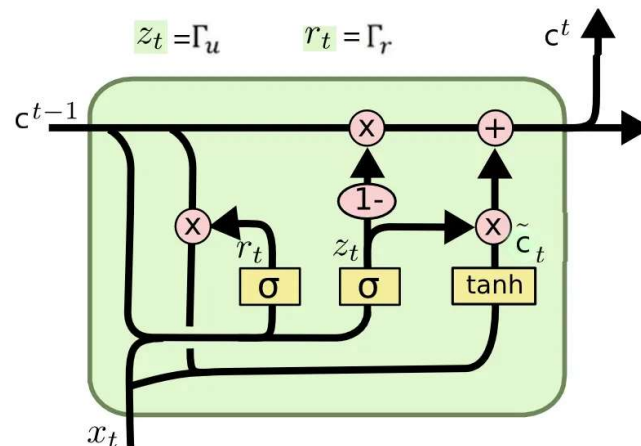


Output gates attenuates to what extent state C gets passed using current output h .

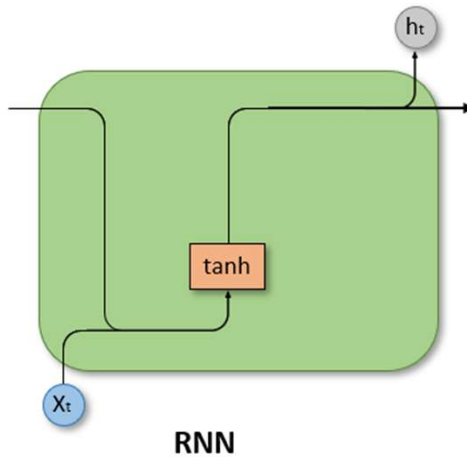
Source: <http://colah.github.io/posts/2015-08-Understanding-LSTMs>

Gated Recurrent Unit (GRU)

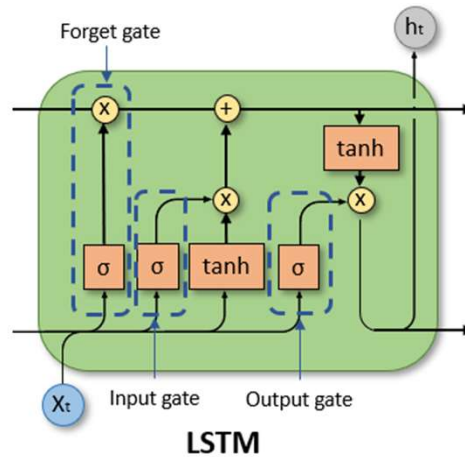
- **Simpler implementation of the LSTM**
 - The GRU model does not have an internal state variable to store memory (a value)
 - Combine **C** and **h** into a single state/output.
 - Combine forget and input gates into update gate, **z**
- **Training time is shorter**
- **Requires less data to capture data properties**



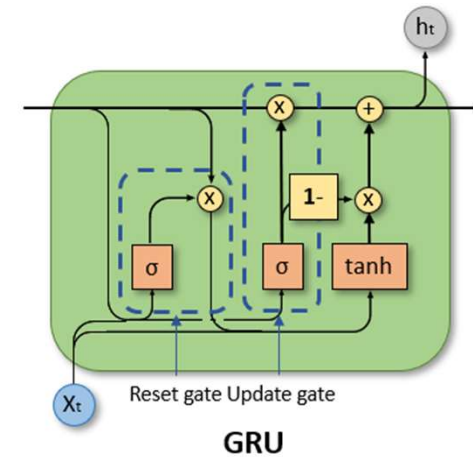
Comparison



- RNNs suffer from the **vanishing gradient** problem. As sequences grow longer they struggle to remember information from earlier steps.
- Less effective for tasks that need understanding of long-term dependencies like machine translation or speech recognition.
- To resolve these challenges more advanced models such as LSTM networks were developed.



- LSTMs are more complex than RNNs which makes them **slower to train** and demands **more memory**.
- LSTMs handle **longer sequences** better they still face challenges with very long-range dependencies.
- LSTM sequential nature also limits the ability to **process data in parallel** which slows down training.



- GRUs are **simpler** and **faster** than LSTMs but they still rely on sequential processing which limits parallelization and **slows training** on long sequences.
- Like LSTMs, they can struggle with very long-range dependencies, in some cases.

Source: [geeksforgeeks.org/deep-learning/rnn-vs-lstm-vs-gru-vs-transformers](https://www.geeksforgeeks.org/deep-learning/rnn-vs-lstm-vs-gru-vs-transformers)

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's steel structure and suspension cables are prominent. The water of the Hudson River is visible in the lower foreground. The sky is a clear, pale blue.

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